

PORT ARANSAS 32 NNE, TX, USA

WMO: 747470

Lat: 28.3045N

Lon: 96.8230W

Elev: 5

StdP: 101.27

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 23906

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.3	(c) 0.2	(d) -6.2	(e) 2.2	(f) 3.0	(g) -4.2	(h) 2.7	(i) 5.4	(j) 5.5	(k) 15.0	(l) 4.8	(m) 13.9	(n) 0.6	(o) N/A	(p) 0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.2	(c) 33.9	(d) 25.8	(e) 33.1	(f) 26.0	(g) 32.5	(h) 26.0	(i) 27.5	(j) 31.3	(k) 27.1	(l) 30.9	(m) 26.8	(n) 30.6	(o) 2.4	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.0	29.1	26.2	21.6	29.1	25.8	21.1	28.9	86.6	31.5	85.2	31.1	84.0	30.8	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.6	(b) 3.9	(c) 3.6	(e) -6.7	(f) 36.6	(g) 2.0	(h) 1.8	(i) -8.2	(j) 37.9	(k) -9.4	(l) 39.0	(m) -10.5	(n) 40.0	(o) -12.0	(p) 41.3		
(4) 4.6	(5) 3.9	(6) 3.6	(7) -7.2	(8) 28.1	(9) 2.3	(10) 0.5	(11) -8.8	(12) 28.4	(13) -10.2	(14) 28.7	(15) -11.5	(16) 29.0	(17) -13.1	(18) 29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	21.3	11.9	15.0	18.1	21.3	24.8	27.8	28.1	28.3	26.6	22.3	17.2	13.8	(6)
(7)		DBStd	6.93	5.20	5.44	4.66	3.63	2.87	1.59	1.27	1.38	2.11	4.36	5.37	5.31	(7)
(8)		HDD10.0	93	41	17	4	0	0	0	0	0	0	1	8	22	(8)
(9)		HDD18.3	667	206	117	63	17	2	0	0	0	0	17	84	161	(9)
(10)		CDD10.0	4216	101	158	254	340	459	533	561	567	499	382	224	139	(10)
(11)		CDD18.3	1748	8	23	55	107	203	283	303	308	249	140	49	19	(11)
(12)		CDH23.3	18056	22	54	158	549	1821	3333	3748	3837	2727	1441	304	61	(12)
(13)	CDH26.7	6652	1	3	11	67	404	1304	1609	1769	1059	388	36	1	(13)	
(14)	Wind	WSAvg	1.6	1.6	1.8	2.1	2.2	2.1	1.8	1.6	1.4	1.2	1.3	1.3	1.4	(14)
(15)	Precipitation	PrecAvg	954	66	55	70	61	91	70	103	82	133	101	66	60	(15)
(16)		PrecMax	1395	240	225	287	147	322	218	509	193	391	281	173	169	(16)
(17)		PrecMin	407	0	2	0	12	0	0	0	15	25	0	0	16	(17)
(18)		PrecStd	280	55	55	67	40	86	62	131	55	92	79	62	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.4	26.7	28.1	31.1	31.7	34.9	34.1	35.3	34.8	31.8	29.1	26.3	(19)
(20)			MCWB	18.3	19.6	19.9	19.0	23.4	25.3	25.5	25.1	24.1	24.6	23.9	21.6	(20)
(21)		2%	DB	23.7	25.0	26.2	28.2	30.5	32.8	33.3	34.0	32.9	30.8	27.8	25.1	(21)
(22)			MCWB	19.0	20.6	20.9	22.1	24.6	25.6	25.9	26.2	25.5	25.2	23.2	20.8	(22)
(23)		5%	DB	22.1	23.6	25.1	27.0	29.7	31.9	32.7	33.3	32.1	30.0	26.5	23.7	(23)
(24)			MCWB	18.1	20.0	20.8	22.4	24.7	26.0	26.1	26.3	25.7	24.7	22.3	20.5	(24)
(25)		10%	DB	20.6	22.1	24.0	26.1	29.0	31.3	32.0	32.7	31.3	29.0	25.1	22.2	(25)
(26)			MCWB	17.8	19.6	20.4	22.1	24.4	25.9	26.1	26.3	25.6	23.8	22.1	20.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	22.4	22.9	25.2	26.4	27.6	27.8	28.0	27.6	26.9	25.3	23.2	(27)
(28)			MCDWB	23.0	25.2	25.8	27.1	29.2	30.9	31.7	32.0	31.2	29.6	27.6	24.6	(28)
(29)		2%	WB	20.6	21.7	22.3	24.4	26.0	27.0	27.3	27.5	27.1	26.1	24.4	22.5	(29)
(30)			MCDWB	22.3	23.9	24.7	26.5	28.7	30.4	31.0	31.7	30.8	28.7	26.5	24.0	(30)
(31)		5%	WB	19.8	21.0	21.8	23.8	25.6	26.8	26.9	27.2	26.7	25.6	23.6	21.6	(31)
(32)			MCDWB	21.3	22.8	23.9	25.8	28.2	30.2	30.7	31.3	30.4	28.5	25.7	23.0	(32)
(33)		10%	WB	18.6	20.4	21.2	23.2	25.2	26.5	26.7	26.8	26.3	25.1	22.6	20.6	(33)
(34)			MCDWB	20.1	21.8	23.1	25.1	27.8	29.9	30.5	30.9	29.9	28.0	24.8	21.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.0	10.2	9.2	8.9	7.7	7.6	8.3	9.2	9.2	11.4	11.7	11.2	(35)
(36)			MCDBR	13.1	10.0	8.5	8.6	8.0	8.1	9.3	10.1	10.3	9.7	10.4	11.1	(36)
(37)			MCWBR	8.8	6.5	4.6	4.8	3.7	3.1	3.8	3.9	4.5	5.3	6.5	7.8	(37)
(38)		5% WB	MCDBR	9.0	8.1	7.6	6.2	5.5	6.4	7.8	8.6	7.8	8.8	8.8	10.0	(38)
(39)			MCWBR	7.1	5.7	4.5	3.9	2.9	2.9	3.5	3.7	3.7	5.5	6.3	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.357	0.367	0.384	0.437	0.482	0.456	0.477	0.454	0.444	0.392	0.380	0.346	(40)	
(41)		taud	2.488	2.463	2.409	2.267	2.173	2.309	2.266	2.333	2.323	2.461	2.443	2.534	(41)	
(42)		Ebn at Noon	884	904	906	863	821	838	819	837	836	865	849	880	(42)	
(43)		Edh at Noon	93	103	116	137	151	131	137	127	125	102	97	86	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.96	3.47	4.55	5.41	6.12	6.71	6.58	6.28	5.22	4.53	3.40	2.68	(44)	
(45)		RadStd	0.33	0.56	0.44	0.50	0.41	0.38	0.58	0.34	0.46	0.47	0.37	0.34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+95

Nomenclature: See separate page